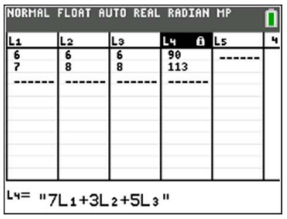
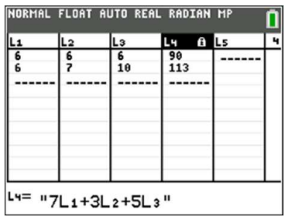
Raffles Institution
H2 Mathematics (9758)
Solution for 2025 A-Level Paper 2

Section A: Pure Mathematics

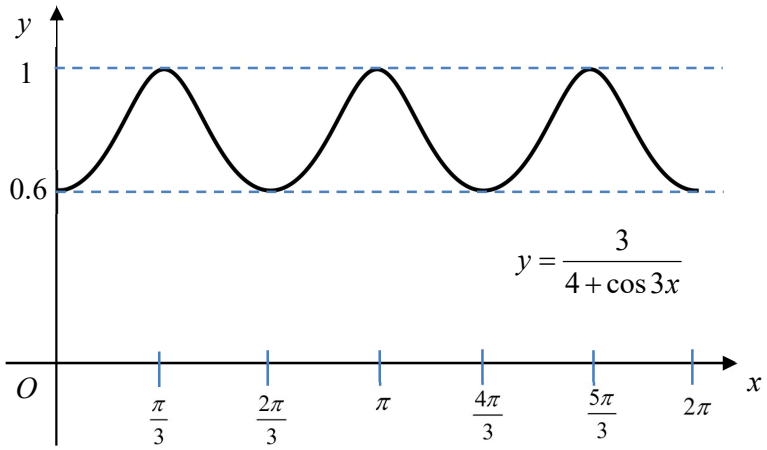
Question 1

No.	Suggested Solution	Remarks for Student
(a)	$8c+11t+5b=114$ --- (1) $5c+14t+7b=112$ --- (2) $9c+9t+4b=110$ --- (3) Using GC, $c=7,\ t=3,\ b=5$	
(b)	Since Sam is second, his score is 113. Sam has at least 6 of each type of token. <u>Method 1</u>: Let the number of cars, trains and boats Sam collects be $x+6, y+6$ and $z+6$ resepectively where $x\geq 0, y\geq 0, z\geq 0$ and $z>y$. Then $7(x+6)+3(y+6)+5(z+6)=113$ $\Rightarrow 7x+3y+5z=23$ By observation, $x=0, y=1, z=4$. Since there is only one solution, Sam had 6 cars, 7 trains and 10 boats. <u>Method 2</u>: Using GC (***Trial and Error***) No of cars | No of trains | No of boats | Score 7 | 8 | 8 | 113 6 | 7 | 10 | 113 But Sam has different numbers of each type of token, and more boats than trains. Since there is only one solution, Sam had 6 cars, 7 trains and 10 boats.	  Notice that L4 is a **formula** (*in inverted commas*) which means that any change in L1 or L2 or L3 will effect a change in computation in L4.

Question 2

No.	Suggested Solution	Remarks for Student
(a)	Given $\overrightarrow{OP}=\begin{pmatrix}-2\\3\\0\end{pmatrix}, \overrightarrow{OQ}=\begin{pmatrix}1\\6\\6\end{pmatrix}$. $3\overrightarrow{QR}=4\overrightarrow{PQ}$ $3\left(\overrightarrow{OR}-\overrightarrow{OQ}\right)=4\left(\overrightarrow{OQ}-\overrightarrow{OP}\right)$ $3\overrightarrow{OR}=7\overrightarrow{OQ}-4\overrightarrow{OP}$ $=7\begin{pmatrix}1\\6\\6\end{pmatrix}-4\begin{pmatrix}-2\\3\\0\end{pmatrix}=\begin{pmatrix}15\\30\\42\end{pmatrix}$ $\therefore \overrightarrow{OR}=\begin{pmatrix}5\\10\\14\end{pmatrix}$	
(b)	$\overrightarrow{PQ}=\begin{pmatrix}1\\6\\6\end{pmatrix}-\begin{pmatrix}-2\\3\\0\end{pmatrix}=\begin{pmatrix}3\\3\\6\end{pmatrix}$ and $\overrightarrow{SQ}=\begin{pmatrix}1\\6\\6\end{pmatrix}-\begin{pmatrix}-1\\-2\\c\end{pmatrix}=\begin{pmatrix}2\\8\\6-c\end{pmatrix}$ Since $\overrightarrow{SQ}$ is perpendicular to $\overrightarrow{PQ}$, $\overrightarrow{SQ}\bullet\overrightarrow{PQ}=0$ $\begin{pmatrix}2\\8\\6-c\end{pmatrix}\bullet\begin{pmatrix}3\\3\\6\end{pmatrix}=0$ $6+24+36-6c=0$ $c=11$	
(c)	$\overrightarrow{PS}=\begin{pmatrix}-1\\-2\\11\end{pmatrix}-\begin{pmatrix}-2\\3\\0\end{pmatrix}=\begin{pmatrix}1\\-5\\11\end{pmatrix}$ Let θ be the angle between $\overrightarrow{PS}$ and $\overrightarrow{PQ}$. $\cos\theta=\dfrac{\begin{pmatrix}1\\-5\\11\end{pmatrix}\cdot\begin{pmatrix}3\\3\\6\end{pmatrix}}{\sqrt{1+25+121}\sqrt{9+9+36}}=\dfrac{54}{\sqrt{7938}}$ $\theta=52.7^\circ$ (1 d.p)	

Question 3

No.	Suggested Solution	Remarks for Student
(a)	$\dfrac{dy}{dx} = y^2 \sin 3x$ $\dfrac{1}{y^2}\dfrac{dy}{dx} = \sin 3x$ Integrate w.r.t x, $\displaystyle\int \frac{1}{y^2}\, dy = \int \sin 3x \, dx$ $-\dfrac{1}{y} = -\dfrac{1}{3}\cos 3x + C$, where $C \in \mathbb{R}$ $\dfrac{1}{y} = \dfrac{1}{3}\cos 3x - C$ When $x = \pi, y = 1$, $1 = \dfrac{1}{3}\cos 3\pi - C$ $C = -\dfrac{1}{3} - 1 = -\dfrac{4}{3}$ $\therefore \dfrac{1}{y} = \dfrac{1}{3}\cos 3x + \dfrac{4}{3} = \dfrac{\cos 3x + 4}{3}$ $y = \dfrac{3}{4 + \cos 3x}$ (shown) where $A = 3, B = 4$.	Remember to include the arbitrary constant , C!
(b) **(i)**	 $\therefore c = \dfrac{3}{5}$ or 1	
(b) **(ii)**	$x = \dfrac{n\pi}{3}$, where $n \in \mathbb{Z}$	

(b) **(iii)**	$f(x)=\dfrac{3}{4+\cos 3x}$ <u>Method 1: Addition formula</u> $f(x+\pi)=\dfrac{3}{4+\cos 3(x+\pi)}$ $=\dfrac{3}{4+\cos(3x+3\pi)}$ $=\dfrac{3}{4+\cos 3x\cos 3\pi-\sin 3x\sin 3\pi}$ $=\dfrac{3}{4-\cos 3x}$ <u>Method 2: Trigonometry properties</u> $f(x+\pi)=\dfrac{3}{4+\cos(3x+3\pi)}$ $=\dfrac{3}{4+\cos(3x+\pi)}$ since $\cos(A+2\pi)=\cos A$ $=\dfrac{3}{4-\cos(3x)}$ since $\cos(A+\pi)=-\cos A$	
(c)	$\dfrac{dy}{dx}=y^2\sin 3x$ $\dfrac{d^2y}{dx^2}=y^2(3\cos 3x)+\left(2y\dfrac{dy}{dx}\right)\sin 3x$ $=(3\cos 3x)y^2+2y\left(y^2\sin 3x\right)\sin 3x$ $=(3\cos 3x)y^2+(2\sin^2 3x)y^3$ where $P=3\cos 3x$ and $Q=2\sin^2 3x$ (Shown)	

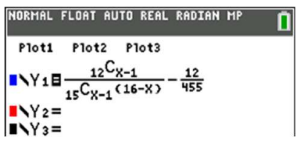
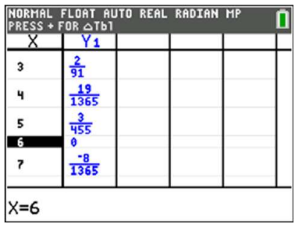
Question 4

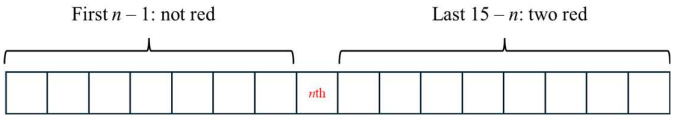
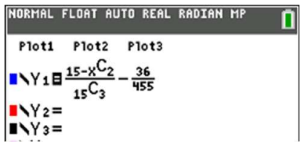
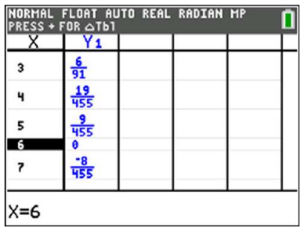
No.	Suggested Solution	Remarks for Student
(a)	Given $V = \frac{1}{3}\pi h^2(45-h)$. When the bowl is full, $h = 15$. $V = \frac{1}{3}\pi(15^2)(45-15) = 2250\pi$ $\therefore$ Time taken to fill the bowl $= \frac{2250\pi}{10\pi} = 225$ seconds	
(b)	$V = \frac{1}{3}\pi h^2(45-h)$ $= \frac{1}{3}\pi(45h^2 - h^3)$ Differentiate w.r.t h, $\frac{dV}{dh} = \frac{1}{3}\pi(90h - 3h^2) = \pi h(30-h)$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{1}{\pi h(30-h)} \times 10\pi$ $= \frac{10}{h(30-h)}$ When $h = 12$ cm, $\frac{dh}{dt} = \frac{10}{12(18)} = \frac{5}{108}$ cm s^{-1} $\therefore$ Depth of water is increasing at $\frac{5}{108}$ cm s^{-1}.	Rewrite V as $\frac{1}{3}\pi(45h^2 - h^3)$ to make it easier to differentiate!
(c)	$\frac{dV}{dt} = 10\pi t$ Integrate w.r.t t, $V = \int 10\pi t \, dt$ $= 5\pi t^2 + C, \quad C \in \mathbb{R}$ When $t = 0$, $V = 0$. So $C = 0$. $\therefore V = 5\pi t^2$. When $V = 972\pi = 5\pi t^2$, $t^2 = \frac{972}{5}$ Since $t \geq 0$, $t = \sqrt{\frac{972}{5}} \approx 13.943 = 13.9$ seconds (3 s.f)	

(d)	When $V = 972\pi$, $V = \frac{1}{3}\pi h^2(45-h) = 972\pi$ $h^2(45-h) = 2916$ From GC, $h = 43.456,\ 9$ or -7.4558 Since $0 \le h \le 15,\ h = 9$. $\frac{dh}{dt} = \frac{1}{\pi h(30-h)} \times \frac{dV}{dt}$ from part **(b)** $= \frac{1}{\pi h(30-h)} \times 10\pi t$ $= \frac{10t}{h(30-h)}$ When $h = 9$, $\frac{dh}{dt} = \frac{10(13.943)}{9(21)} = 0.738$ cm s^{-1}. (3 s.f) $\therefore$ Depth of water is increasing at 0.738 cm s^{-1}.	

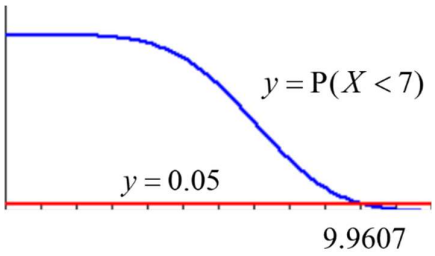
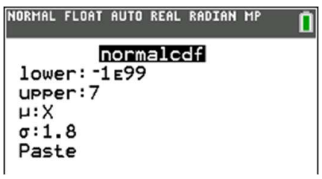
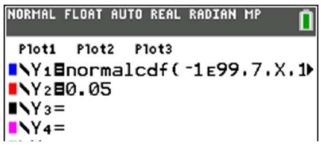
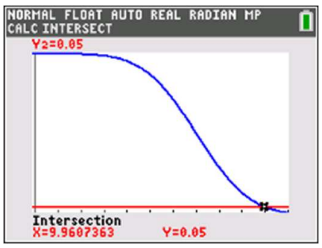
Section B: Probability and Statistics

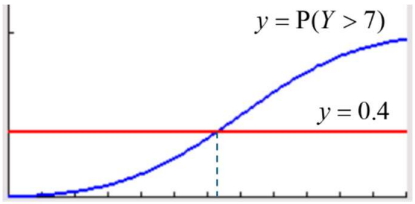
Question 5

No.	Suggested Solution	Remarks for Student
(a)	Method 1: Direct method, consider individual colours in second stage Probability that the 2 counters are of different colours $=$ P(red and blue) + P(red and yellow)+ P(blue and yellow) $=\frac{3}{15}\times\frac{5}{14}\times 2!\ +\frac{3}{15}\times\frac{7}{14}\times 2!\ +\frac{5}{15}\times\frac{7}{14}\times 2!$ $=\frac{71}{105}$ **OR** $=\frac{{}^{3}C_{1}\,{}^{5}C_{1}+{}^{3}C_{1}\,{}^{7}C_{1}+{}^{5}C_{1}\,{}^{7}C_{1}}{{}^{15}C_{2}}=\frac{71}{105}$ Method 2: Direct method, group colours in second stage Probability that the 2 counters are of different colours $=$ P(red,not red) + P(blue, not blue) + P(yellow, not yellow) $=\frac{3}{15}\times\frac{12}{14}+\frac{5}{15}\times\frac{10}{14}+\frac{7}{15}\times\frac{8}{14}$ $=\frac{71}{105}$ Method 3: Complement Method Probability that the 2 counters are of different colours $=1-[$ P(2 red) + P(2 blue) + P(2 yellow) $]$ $=1-\left(\frac{3}{15}\times\frac{2}{14}+\frac{5}{15}\times\frac{4}{14}+\frac{7}{15}\times\frac{6}{14}\right)$ $=\frac{71}{105}$	While a certain method may be more intuitive for you, do read all the possible solutions here and ensure that you understand all of them as all the methods are "equally" accessible.
(b)	Method 1: Selection of the first $n-1$ counters P(first $n-1$ not red and nth red) $=\frac{36}{455}$ P(first $n-1$ not red) $\cdot$ P(nth red) $=\frac{36}{455}$ $\frac{{}^{12}C_{n-1}}{{}^{15}C_{n-1}}\times\frac{3}{15-(n-1)}=\frac{36}{455}$ $\frac{{}^{12}C_{n-1}}{{}^{15}C_{n-1}(16-n)}-\frac{12}{455}=0$ Using GC, from the table of values, $n=6$	GC screenshots:  

	<u>Method 2: Consider the arrangement of the last 15– n counters</u>  P(2 red in the last $15-n$ positions and red at the nth position) $= \dfrac{^{15-n}C_2}{^{15}C_3} = \dfrac{36}{455}$ Consider $\dfrac{^{15-n}C_2}{^{15}C_3} - \dfrac{36}{455} = 0$. Using GC, from the table of values, $n = 6$ **OR** $\dfrac{^{15-n}C_2}{^{15}C_3} = \dfrac{36}{455}$ $^{15-n}C_2 = 36$ $\dfrac{(15-n)(14-n)}{2} = 36$ $n^2 - 29n + 138 = 0$ $(n-6)(n-23) = 0$ $n = 6$ or $n = 23$ (reject since n is at most 15) <u>Method 3: Consider the arrangement of the first n counters</u> $\dfrac{^{12}P_{n-1} \times 3}{^{15}P_n} = \dfrac{36}{455}$ $\dfrac{^{12}P_{n-1}}{^{15}P_n} - \dfrac{12}{455} = 0$ Using GC, from the table of values, $n = 6$	GC screenshots:  

Question 6

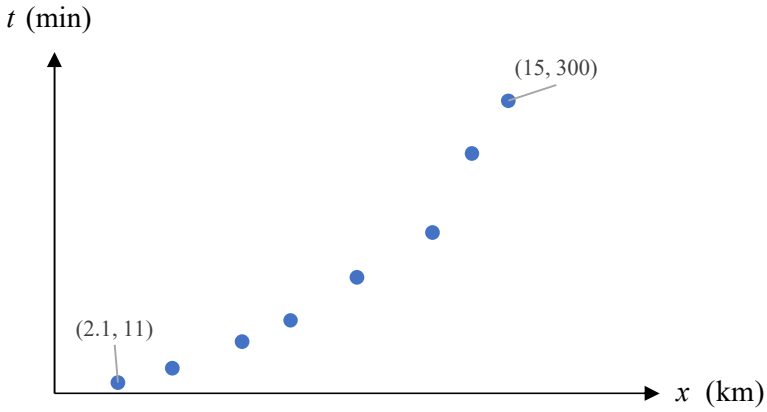
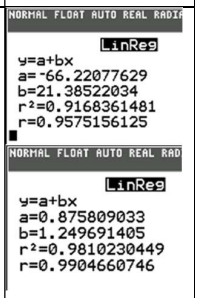
No.	Suggested Solution	Remarks for Student
(a)	$\mathrm{P}(X < 7) = 0.05$	
(b)	Given $X \sim \mathrm{N}(\mu, 1.8^2)$. <u>Method 1: Standardisation</u> $\mathrm{P}(X < 7) = 0.05$ $\mathrm{P}\left(Z < \dfrac{7-\mu}{1.8}\right) = 0.05$ From GC, $\dfrac{7-\mu}{1.8} = -1.6449$ $\mu \approx 9.9607 = 9.96$ (3 s.f) <u>Method 2: Graphical method</u>  $\mathrm{P}(X < 7) = 0.05$ Using GC, $\mu = 9.96$ (3 s.f)	Consider using GC method as a check :   
(c)	Given $Y \sim \mathrm{N}(\lambda, 2.8^2)$. $\mathrm{P}((X > 7) \cap (Y > 7)) = 0.38$ Since X and Y are independent, $\mathrm{P}(X > 7) \times \mathrm{P}(Y > 7) = 0.38$ $0.95\mathrm{P}(Y > 7) = 0.38$ $\mathrm{P}(Y > 7) = 0.4$ <u>Method 1: Standardisation</u> $\mathrm{P}(Y > 7) = 0.4$ $\mathrm{P}\left(Z > \dfrac{7-\lambda}{2.8}\right) = 0.4$ From GC, $\dfrac{7-\lambda}{2.8} = 0.25335$ $\lambda \approx 6.2906 = 6.29$ (3 s.f)	Similar to **(b)**, consider using GC method to check your answer.

	Method 2: Graphical method  6.2906 $\mathrm{P}(Y > 7) = 0.4$ Using GC, $\lambda = 6.29$ (3 s.f)	
(d)	$\mathrm{P}(Y > a) = 2\mathrm{P}(Y > 7) = 0.8$ From GC, $\mathrm{P}(Y > 3.93) = 0.8$ $\therefore a = 3.93$	

Question 7

No.	Suggested Solution
(a)	Since there are 8 cards and 2 "3" cards and 2 "7" cards, number of different arrangements $= \dfrac{8!}{2!2!} = 10080$
(b)	<u>Method 1: Direct (Preferred)</u> Number of different arrangements $= \underbrace{{}^3C_2 \times 2!}_{\substack{\text{Choose and arrange}\\ \text{2 even numbers}\\ \text{for the start and end}}} \times \underbrace{{}^4C_1}_{\substack{\text{Choose 1 out of}\\ \text{4 middle spaces for}\\ \text{the last even number}}} \times \underbrace{\dfrac{5!}{2!2!}}_{\substack{\text{arrange the}\\ \text{rest of the}\\ \text{odd numbers}}} = 720$ <u>Method 2: Complement</u> Number of different arrangements $= {}^3C_2 \times 2! \times \left[\dfrac{6!}{2!2!} - 2 \times \dfrac{5!}{2!2!}\right] = 720$
(c)	Number of different arrangements $= \underbrace{\dfrac{6!}{2!2!}}_{\substack{\text{Number}\\ \text{of units is 6}}} \times \underbrace{3!}_{\substack{\text{arrange the}\\ \text{even numbers}}} = 1080$
(d)	<u>Method 1: Reasoning</u> For all arrangements of the 5 odd digits, there are 6 possible slots for the group of 3 even digits. There are 2 of the 6 slots (leftmost or rightmost) in which all the odd digits will also be together. Required probability $= \dfrac{2}{6} = \dfrac{1}{3}$ <u>Method 2: Formula</u> $\text{P(odd digits all together} \mid \text{even digits all together)}$ $= \dfrac{\text{P(odd digits all together and even digits all together)}}{\text{P(even digits all together)}}$ $= \dfrac{\text{n(odd digits all together and even digits all together)}}{\text{n(even digits all together)}}$ $= \dfrac{2! \times 3! \times \dfrac{5!}{2!2!}}{1080} = \dfrac{360}{1080} = \dfrac{1}{3}$ Note : Number of different arrangements when even digits are all together and odd digits are also all together $= \underbrace{2!}_{\substack{\text{Number}\\ \text{of units is 2}}} \times \underbrace{3!}_{\substack{\text{arrange the}\\ \text{even numbers}}} \times \underbrace{\dfrac{5!}{2!2!}}_{\substack{\text{arrange the}\\ \text{even numbers}}} = 360$

Question 8

No.	Suggested Solution	Remarks for Student
(a)	The data points do not lie close to a straight line and the correlation coefficient is 0.39 which is not close to 1. Therefore, there is a weak positive linear relationship between best performance in high jump and long jump amongst athletes taking part in both events.	
(b)(i)	 The scatter plot suggests that as x increases, t increases at an increasing rate.	Always do a quick count to ensure that you have the correct number of points! 8 in this case. Label the mininum and maximum values for the 2 variables too!
(b)(ii) **(A)**	For $t = ax + b$, $r \approx 0.958$ For $t = cx^2 + d$, $r \approx 0.990$ Since the value of r for the model $t = cx^2 + d$ is closer to 1, $t = cx^2 + d$ is a better fit to the data.	
(b)(ii) **(B)**	$t \approx 1.2497x^2 + 0.87581$ $t = 1.25x^2 + 0.876$ (3 s.f)	
(b)(iii) **(A)**	When $x = 11$, $t = 1.2496(11)^2 + 0.87581 = 152$ (3 s.f) The estimated time Dilip would take for a run of 11 km is 152 minutes.	
(b)(iii) **(B)**	The estimate is reliable since the value $x = 11$ is within the data range [2.1, 15] and $r \approx 0.990$ is very close to 1, suggesting a strong positive linear correlation between t and x^2.	

Question 9

No.	Suggested Solution	Remarks for Student
(a)	The **event** that a refrigerator is faulty is independent of the **event** that another refrigerator is faulty. The probability of a refrigerator being faulty remains constant.	
(b)	$X \sim \mathrm{B}(90, 0.02)$ $\mathrm{P}(X > 1)$ $= 1 - \mathrm{P}(X \leq 1)$ ≈ 0.53957 $= 0.540$ (3 s.f)	
(c)	Let W be the number of days, out of 5 where Alan finds more than one faulty refrigerator. $W \sim \mathrm{B}(5, \mathrm{P}(X > 1))$ $W \sim \mathrm{B}(5, 0.53957)$ $\mathrm{P}(W < 2) = \mathrm{P}(W \leq 1) = 0.142$ (3 s.f)	Remember to use at least 5 s.f for $\mathrm{P}(X > 1)$ so that the accuracy for the final answer will not be compromised.
(d)	Let T be the number of refrigerator, out of 450, that are faulty. $T \sim \mathrm{B}(450, 0.02)$ $\mathrm{P}(T < 10) = \mathrm{P}(T \leq 9) = 0.587$ (3 s.f)	
(e)	$Y \sim \mathrm{B}(60, 0.03)$ and $X \sim \mathrm{B}(90, 0.02)$ where X and Y are independent. Required probability $= \mathrm{P}(X = 0)\mathrm{P}(Y = 2) + \mathrm{P}(X = 1)\mathrm{P}(Y = 1) + \mathrm{P}(X = 2)\mathrm{P}(Y = 0)$ $\approx 0.1766878643 = 0.177$ (3 s.f)	Note: The answer 0.1766878643 is meant for you to check that you are on the right track. There is no need to write this during the exams.

Question 10

No.	Suggested Solution	Remarks for Student
(a)	Since the two machines are different and their productions are not affecting each other, it is reasonable to assume that the two distributions are independent.	
(b)	Let P and Q be the mass, in kg, of a packet of flour produced Machine P and Q respectively. $P \sim \mathrm{N}(2.2\ 0.1^2)$ and $Q \sim \mathrm{N}(2.1.\ 0.05^2)$ $\mathrm{E}(P-Q) = \mathrm{E}(P) - \mathrm{E}(Q) = 2.2 - 2.1 = 0.1$ $\mathrm{Var}(P-Q) = \mathrm{Var}(P) + \mathrm{Var}(Q) = 0.1^2 + 0.05^2 = 0.0125$ $P-Q \sim \mathrm{N}(0.1,\ 0.\ 0.0125)$ $\mathrm{P}(P>Q) = \mathrm{P}(P-Q>0) = 0.814$ (3 s.f)	Define the r.v and write down the distribution!
(c)	Let $T = P_1 + P_2 + P_3 + Q_1 + Q_2 + \cdots + Q_5$ $\mathrm{E}(T) = 3\mathrm{E}(P) + 5\mathrm{E}(Q) = 3(2.2) + 5(2.1) = 17.1$ $\mathrm{Var}(T) = 3\mathrm{Var}(P) + 5\mathrm{Var}(Q) = 3(0.1^2) + 5(0.05^2) = 0.0425$ $T \sim \mathrm{N}(17.1, 0.0425)$ $\mathrm{P}(A>17) = 0.686$ (3 s.f)	
(d)	Let μ be the **population mean mass of packets of flour** produced by Machine P, after the adjustment. $\mathrm{H}_0 : \mu = 2.2$ $\mathrm{H}_1 : \mu \neq 2.2$	Remember to define μ, as requested by the question.
(e)	Let $y = x - 2$. Then $\sum y = 4.5$ and $\sum y^2 = 1.11$ Unbiased estimate of the population mean, $\bar{x} = \bar{y} + 2 = 2 + \frac{4.5}{30} = 2.15$ Unbiased estimate of the population variance, $s^2 = \frac{1}{29}\left(1.11 - \frac{(4.5)^2}{30}\right) = 0.015$	

(f)	$H_0 : \mu = 2.2$ $H_1 : \mu \neq 2.2$ Perform a 2-tail test at 5% level of significance Under H_0, since sample size, $n = 30$ is large, by Central Limit Theorem, $\overline{X} \sim N\left(2.2, \frac{0.015}{30}\right)$ approximately. Using z-test, p-value $= 2P\left(\overline{X} \leq 2.15\right) \approx 0.0253$ (3 s.f) Since p-value $= 0.0253 < 0.05$, we reject H_0 and conclude that there is sufficient evidence, at the 5% level of significance, that the **mean** mass of packets of flour produced by Machine *P* after the adjustment differs from 2.2 kg.	Note how the conclusion is crafted. Do not leave out any part of the conclusion.
(g)	After Machine *P* has been adjusted, the masses of packets of flour may no longer follow a normal distribution. If a sample of fewer than 30 packets had been used, Central Limit Theorem cannot be applied. Hence, the normal distribution would not provide a suitable model for the distribution of the sample mean mass of packets of flour.	
(h)	If the sample of 30 packets of flour produced by Machine *P* had not been chosen randomly, the sample may not be representative of all the packets of flour produced by machine *P* after the adjustment due to selection bias.	